

17. Using method of partial fraction find

$$L^{-1}\left[\frac{5s^2 - 15s - 11}{(s+1)(s-2)^3}\right]. \quad (11)$$

Or

18. Using Laplace transform solve $y'' - 3y' + 2y = e^{2t}$,
 $y(0) = -3$ and $y'(0) = 5$. (11)

19. Find the Fourier Transform for $f(x) = \begin{cases} 1, & |x| < a \\ 0, & |x| > a \end{cases}$.

Hence deduce that $\int_0^{\infty} \left(\frac{\sin t}{t}\right) dt = \frac{\pi}{2}$. (11)

Or

20. (a) Evaluate $\int_0^{\infty} \frac{dx}{(x^2 + a^2)^2}$ using Parseval's identity of Fourier cosine transform. (6)
- (b) Find Fourier sine transform of $f(x) = xe^{-ax}$, $a > 0$. (5)

5781007

B.Tech. DEGREE EXAMINATION, SEPTEMBER 2020.

Second Semester

Common to All Branches

MATHEMATICS – II

(From 2013 – 14 onwards)

Time : Three hours

Maximum : 75 marks

PART A — (10 × 2 = 20 marks)

Answer ALL the questions.

All questions carry equal marks.

1. Verify Cayley Hamilton theorem for $A = \begin{bmatrix} 1 & 2 \\ 4 & 5 \end{bmatrix}$.
2. The two Eigen values of a matrix $A = \begin{bmatrix} 4 & 6 & 6 \\ 1 & 3 & 2 \\ -1 & -5 & -2 \end{bmatrix}$ are equal to the double of the third Eigen value. Find Eigen value of A^2 .
3. Show that $\text{Curl grad } \phi = 0$.
4. If S is a closed surface enclosed is the volume V and $\vec{A} = ax\hat{i} + by\hat{j} + cz\hat{k}$. Show that $\iint_S \vec{A} \cdot \hat{n} ds = (a + b + c)V$.

5. Verify Final value theorem on Laplace transform for $f(t) = ae^{-bt}$, $b > 0$.
6. Does $L\left[\frac{\cos at}{t}\right]$ exist?
7. Find $L^{-1}\left[\frac{1}{(s+2)^2}\right]$.
8. State convolution theorem on Laplace inverse transform.
9. State and prove change of scale property of Fourier Transform.
10. Find Fourier Transform for $f(x) = x$, $|x| \leq 1$.

PART B — (5 × 11 = 55 marks)

Answer ALL the Units, choosing ONE full questions from each Unit.

11. Verify Cayley Hamilton theorem for the matrix $A = \begin{bmatrix} 1 & 2 & -1 \\ 2 & -3 & 1 \\ 2 & 4 & -2 \end{bmatrix}$ and hence find A^4 . (11)

Or

12. Diagonalise the matrix $A = \begin{bmatrix} 2 & 2 & 0 \\ 2 & 1 & 1 \\ -7 & 2 & -3 \end{bmatrix}$. (11)
13. Verify Gauss Divergence theorem for $\vec{F} = 4xz\hat{i} - y^2\hat{j} + yz\hat{k}$ where S is the surface of the cube bounded by the lines $x=0, y=0, z=0, x=2, y=2, z=2$. (11)

Or

14. Verify Stoke's theorem for $\vec{A} = x^2\hat{i} + xy\hat{j}$ taken over the square region bounded by the lines $x=0, y=0, x=2, y=2$. (11)
15. Find the Laplace transform of the Half-wave

$$\text{rectifier function } f(t) = \begin{cases} \sin \omega t, & 0 < t < \frac{\pi}{\omega} \\ 0, & \frac{\pi}{\omega} < t < \frac{2\pi}{\omega} \end{cases} \quad \text{where}$$

$$f\left(t + \frac{2\pi}{\omega}\right) = f(t). \quad (11)$$

Or

16. (a) Find $L[t^2 e^{-3t}]$. (5)
- (b) Find $L\left[\frac{1-e^t}{t}\right]$. (6)